

Final Project – From Simulation to Machine Learning

Objective

In this final project, you will complete the **entire pipeline** from detailed EnergyPlus simulation to data-driven prediction.

The goal is not only to build accurate models but also to **understand how physical simulation and machine learning differ** — in data source, scale, precision, and interpretation.

Step 1 – Generate Simulation Data

Use your simplified EnergyPlus model (`model_week4.idf`) to perform a **multi-parameter batch simulation**.

Parameter	Description	Levels
Location	Climate zones representing U.S. cities	San Francisco, Sacramento, Chicago, New York
Wall R-value	Wall insulation level (ft ² ·°F·hr/Btu)	14.3, 21.9, 29.7
WWR	Window-to-Wall Ratio	0.25, 0.40, 0.60
SHGC	Solar Heat Gain Coefficient	0.25, 0.40, 0.60
Occupancy Density	People per floor area	0.04, 0.06, 0.08
Lighting Density	Lighting power density (W/m ²)	7.5, 9
Cooling Setpoint	Indoor cooling temperature (°C)	24, 26, 28

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Total simulation cases: $4 \times 3 \times 3 \times 3 \times 3 \times 2 \times 3 = 1944$ (or more if you add optional parameters)

Save the outputs:

- `results.csv` → cooling energy results
- `all_parameters.csv` → corresponding parameter combinations

Step 2 – Feature Engineering

In this stage, you will **define and construct your own features** that best describe the relationship between building configuration and energy performance.

You may consider — but are not limited to — the following feature types:

Category	Possible Examples	Description
Climate Indicators	Cooling/Heating Degree Days (CDD/HDD), average or extreme temperatures, solar radiation	Capture how external conditions drive cooling demand
Design Variables	R-value, WWR, SHGC, setpoint, lighting/occupancy density	Represent envelope and operation parameters
Derived Features	Ratios, interaction terms (e.g., $WWR \times SHGC$), normalized indicators	Capture nonlinear or combined effects

 There is **no single correct method** — you are encouraged to explore different feature definitions, transformations, or normalization strategies that improve model performance or interpretability.

 **Goal:** create a structured dataset linking simulation parameters and climate factors to building energy response.

Step 3 – Train Machine Learning Models

Using your combined dataset:

1. Define input (x) and target (y) variables.
2. Split your data into training (San Francisco, NYC, Chicago) and testing (Sacramento) sets.

- 3. Train and compare multiple regression models (including at least Linear, Decision Tree, Random Forest).
- 4. Evaluate performance using MSE, RMSE, MAE, and R² metrics.

🟢 **Goal:** test how well data-driven models can approximate simulation-based results.

🔍 Step 4 – Compare and Reflect

After completing your models, analyze and discuss the **differences between simulation and machine learning**.

Your reflection should go beyond numeric accuracy — focus on conceptual distinctions:

Aspect	Simulation (EnergyPlus)	Machine Learning
Data Source	Physics-based, derived from model geometry and materials	Pattern-based, relies on data relationships
Scale	Hourly/sub-hourly, spatially detailed	Aggregated across samples and features
Precision	Controlled by physical fidelity	Limited by data quality and coverage
Interpretability	Physically explicit	Statistical or black-box, depends on algorithm
Computation	Intensive, per simulation run	Lightweight after training
Use Case	Design analysis, calibration, validation	Prediction, optimization, surrogate modeling

🧠 Reflection Question:

How do the **data scale** and **precision** differ between simulation and ML?
What trade-offs arise when replacing physics-based detail with data-driven learning?



Deliverables (required — slides only)

Submit a short slide deck (PDF or PPTX, 4–8 slides). The deck must be self-contained and clearly communicate your workflow, results, and interpretation. Required slide content:

1. Title slide — project title, authors, date (1 slide).
2. Climate distributions — 1 slide with plots showing outdoor temperature distributions for each location (histogram/violin + short summary stats).
3. ML evaluation — 1–2 slides that include:
 - Key metrics (MSE, RMSE, MAE, R^2) for the models you report,
 - Predicted vs actual scatter (with 1:1 line) and a brief note on major errors,
 - One compact feature-importance or interpretability plot.
4. Reflection & conclusions — 1–2 slides that:
 - Compare data-driven (ML) results vs physics-based simulation (strengths, limitations, recommended use cases),
 - State key takeaways and 1–2 recommended next steps.



Submission instructions

- Filename: final_slides.pdf or final_slides.pptx
- Submit slides only; including data/notebooks/code is optional but encouraged (provide a link or archive if available).

Grading (slides-focused, 100 pts)

- Clarity & communication: 25 pts
- Climate & data visualisation: 25 pts
- ML evaluation (metrics + plots): 20 pts
- Reflection & interpretation: 30 pts

Submission Deadline

Submit your final slide deck (final_slides.pdf or final_slides.pptx) via the course LMS by 28/11/2025, 23:59 local time.

If you include links to data/code/notebooks, add them in an appendix slide and ensure the links are accessible.